WALCHAND COLLEGE OF ENGINEERING, SANGLI

WALCHAND LINUX USERS' GROUP

OPEN SOURCE DAY

EXCITING PRIZES

LIMITED SEATS

S1

- Introduction to GIT
- GIT Commands
- GIT Branching
- GITHub Contribution

REGISTER HERE

S2

- Introduction to CyberSecurity
- Various CyberSecurity Tools
- Security Precautions
- Detection and Solutions

CONNECT WITH US

www.wcewlug.org

18 JAN

MAIN CCF

FOR ANY QUERY
9579047160
7219685991

Ms. Pooja Bhosale
President
Walchand Linux Users' Group

Dr. A. J. Umbarkar
Staff Advisor
Walchand Linux Users' Group

Dr. R. R. Rathod
HoD Information Technology

Dr. A. R. Surve
Staff Advisor and HoD Computer Science and Engineering

Dr. M. M. Khot
Dean (SW)
Walchand College of Engineering

Dr. S. S. Rathod
Director
Walchand College of Engineering

WLUG
COMMUNITY | KNOWLEDGE | SHARE

TABLE OF CONTENTS

- Cybersecurity
- CIA Triad
- Cybersecurity Tools

WLUG
COMMUNITY | KNOWLEDGE | SHARE

INTRODUCTION TO CYBERSECURITY

WLUG
COMMUNITY | KNOWLEDGE | SHARE

INTRODUCTION

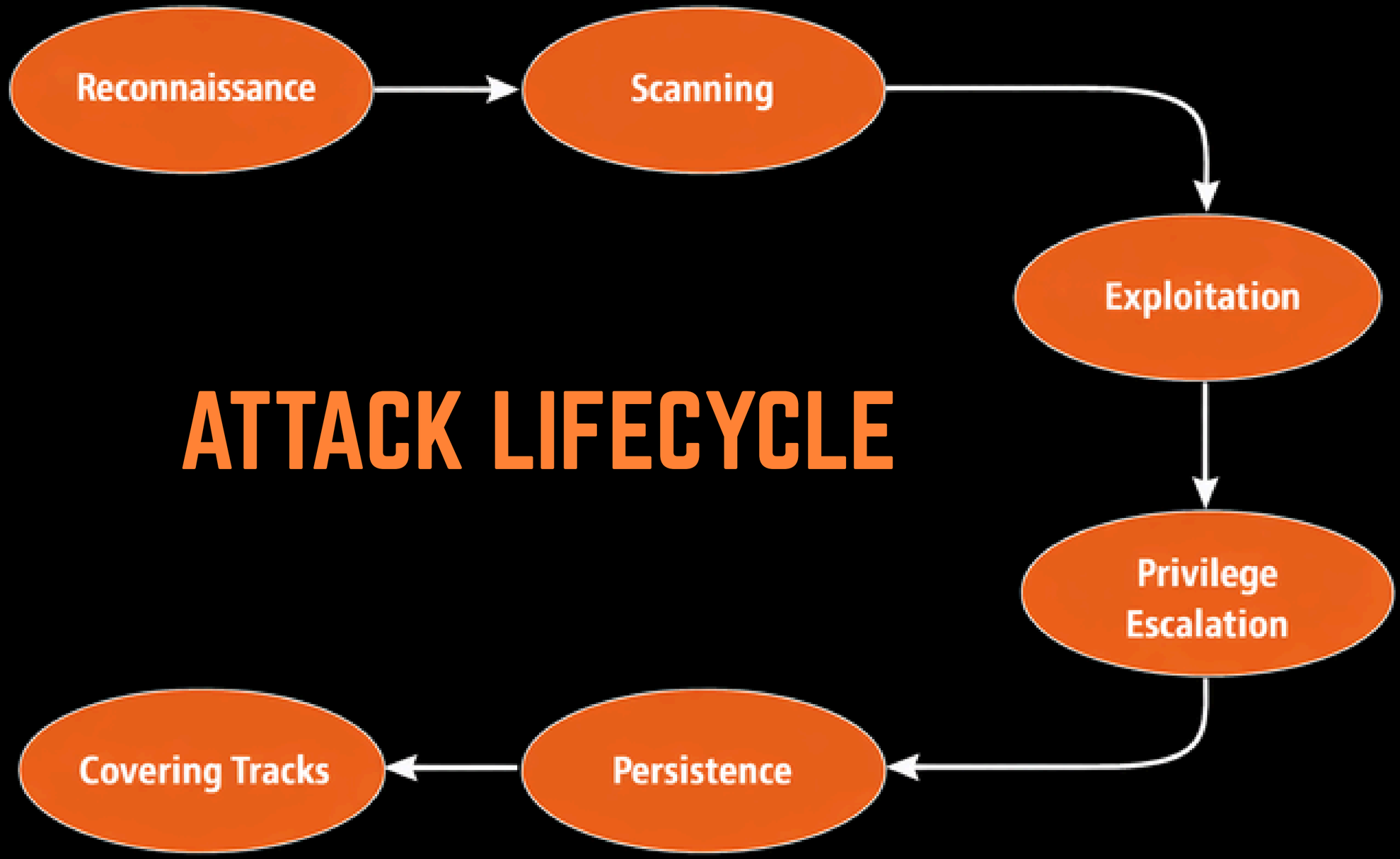
- Cybersecurity is the practice of defending digital systems, networks and data against intentional attacks and accidental failures.

- Cybersecurity is a blend of:
  - Technology
  - Processes
  - People

ATTACK?

- Attacks are intentional actions designed to compromise systems or data
- They aim to disrupt, steal or manipulate digital operations
- Most attacks occur in multiple stages, exploiting weaknesses step by step

ATTACK LIFECYCLE



WLUG
COMMUNITY | KNOWLEDGE | SHARE

TYPES OF ATTACKS

MALWARE

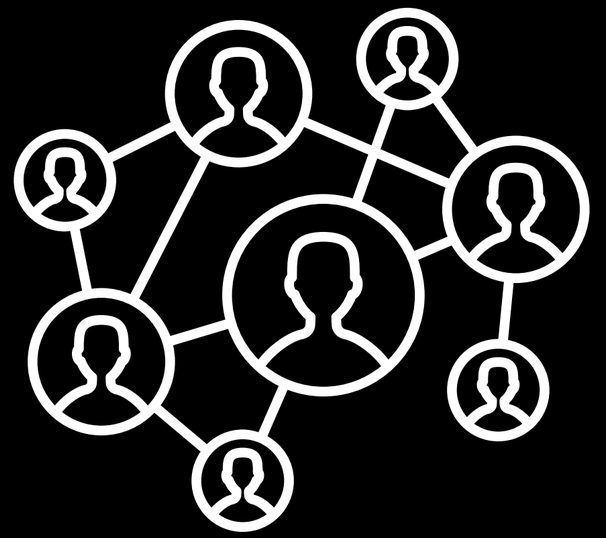
NETWORK

APPLICATION

WLUG
COMMUNITY | KNOWLEDGE | SHARE

NETWORK ATTACK

- A **network attack** is any unauthorized attempt to monitor, manipulate or damage a digital network.
- Some examples of Network Attacks are :
  - **MAN-IN-THE-MIDDLE**
  - **DDoS**



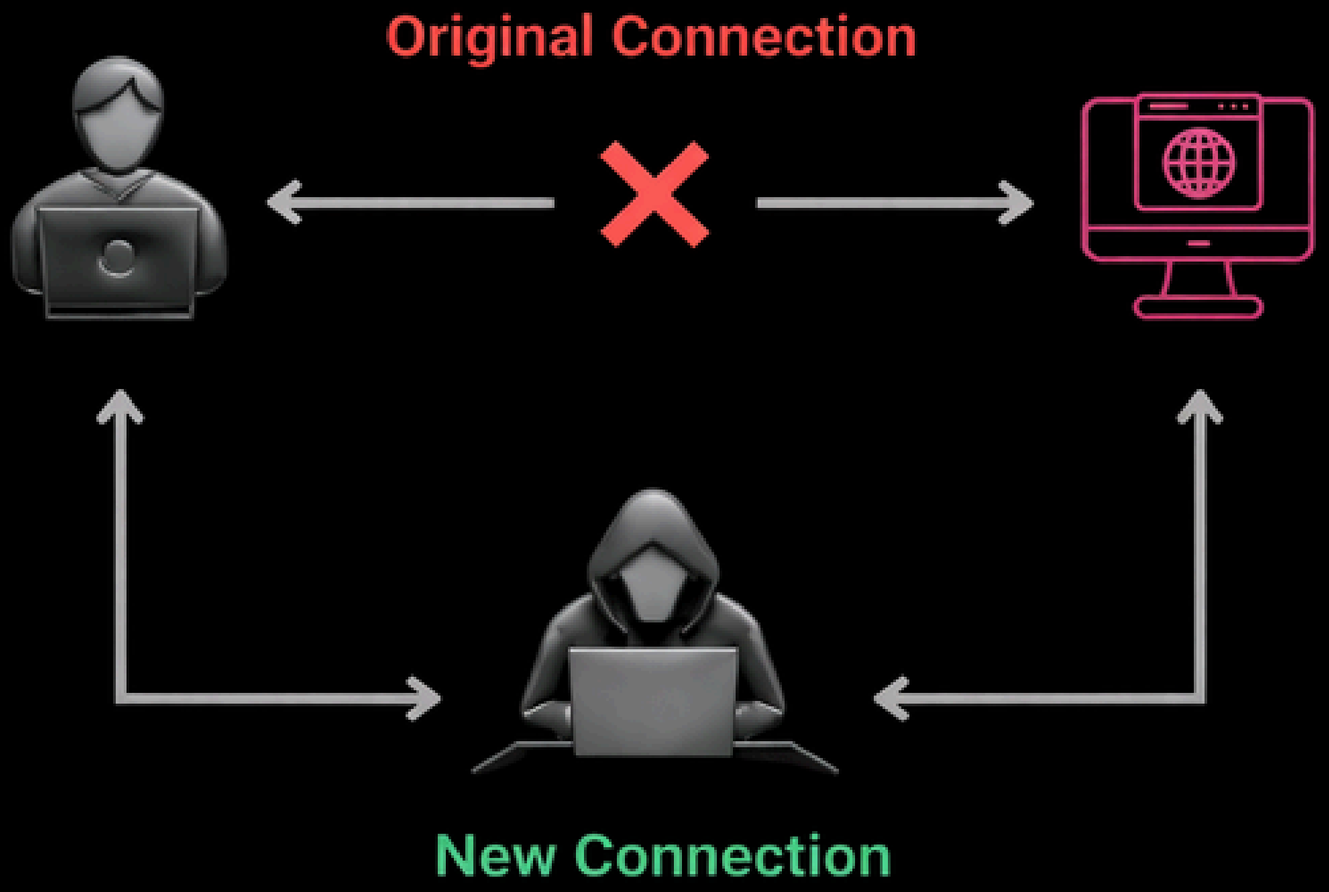
WLUG
COMMUNITY | KNOWLEDGE | SHARE

MAN-IN-THE-MIDDLE

A (MITM) attack is a cyber attack where a hacker secretly intercepts and alters communication between two parties.

WLUG
COMMUNITY | KNOWLEDGE | SHARE

MAN-IN-THE-MIDDLE



WLUG
COMMUNITY | KNOWLEDGE | SHARE

DISTRIBUTED DENIAL OF SERVICE (DDOS)

DDoS attack floods a target server, network or website with overwhelming traffic from BOTNET, making service slow or completely unavailable to legitimate users.

WLUG
COMMUNITY | KNOWLEDGE | SHARE

MALWARE ATTACK

- Virus: Malicious code that attaches to files and needs human action to spread.
- Worm: Self replicating malware that spreads automatically across networks.

WLUG
COMMUNITY | KNOWLEDGE | SHARE

MALWARE ATTACK

- Trojan: Malicious software disguised as legitimate programs to trick users.
- Ransomware: Malware that encrypts your data and demands payment for access.

WLUG
COMMUNITY | KNOWLEDGE | SHARE

APPLICATION ATTACK

- SQL injection
- XSS(Cross Site Scripting)
- Command Injection

WLUG
COMMUNITY | KNOWLEDGE | SHARE

KEEP IN MIND!!

MOST MODERN BREACHES SUCCEED NOT DUE TO ADVANCED HACKING, BUT DUE TO MISCONFIGURATIONS, WEAK PASSWORDS, OR HUMAN ERROR.

WLUG
COMMUNITY | KNOWLEDGE | SHARE

THE CIA TRIAD

- The CIA Triad defines the fundamental goals of cybersecurity
- Every security decision is a trade-off between following listed THREE pillars

WLUG
COMMUNITY | KNOWLEDGE | SHARE

01 CONFIDENTIALITY

- Role-based access control
- Encryption (data at rest & in transit)
- Authentication and authorization

WLUG
COMMUNITY | KNOWLEDGE | SHARE

02 INTEGRITY

- Cryptographic hashing
- Digital signatures
- File integrity monitoring

WLUG
COMMUNITY | KNOWLEDGE | SHARE

03 AVAILIBLITY

- Redundancy and load balancing
- Regular backups
- DDoS mitigation strategies

WLUG
COMMUNITY | KNOWLEDGE | SHARE

TOOLS USED IN CYBERSECURITY

- JOHN THE RIPPER
- HASHCAT
- NMAP
- MALTEGO
- WIRESHARK

WLUG
COMMUNITY | KNOWLEDGE | SHARE

JOHN THE RIPPER

John the Ripper is a password auditing and **recovery tool.**

Core Purpose

- Evaluate password strength
- Identify weak authentication practices

TYPES OF MODES

- Single Crack Mode
- Wordlist (Dictionary) Mode
- Incremental Mode (Brute-Force)

WHAT IS A HASH?

- Hashing is a one-way process that converts data (like passwords or files) into a unique, fixed-length string (a hash) using a hash function.

HASH FUNCTION

- A mathematical algorithm (e.g., SHA-256, MD5) that takes input data of any size and produces a fixed-length hash value.

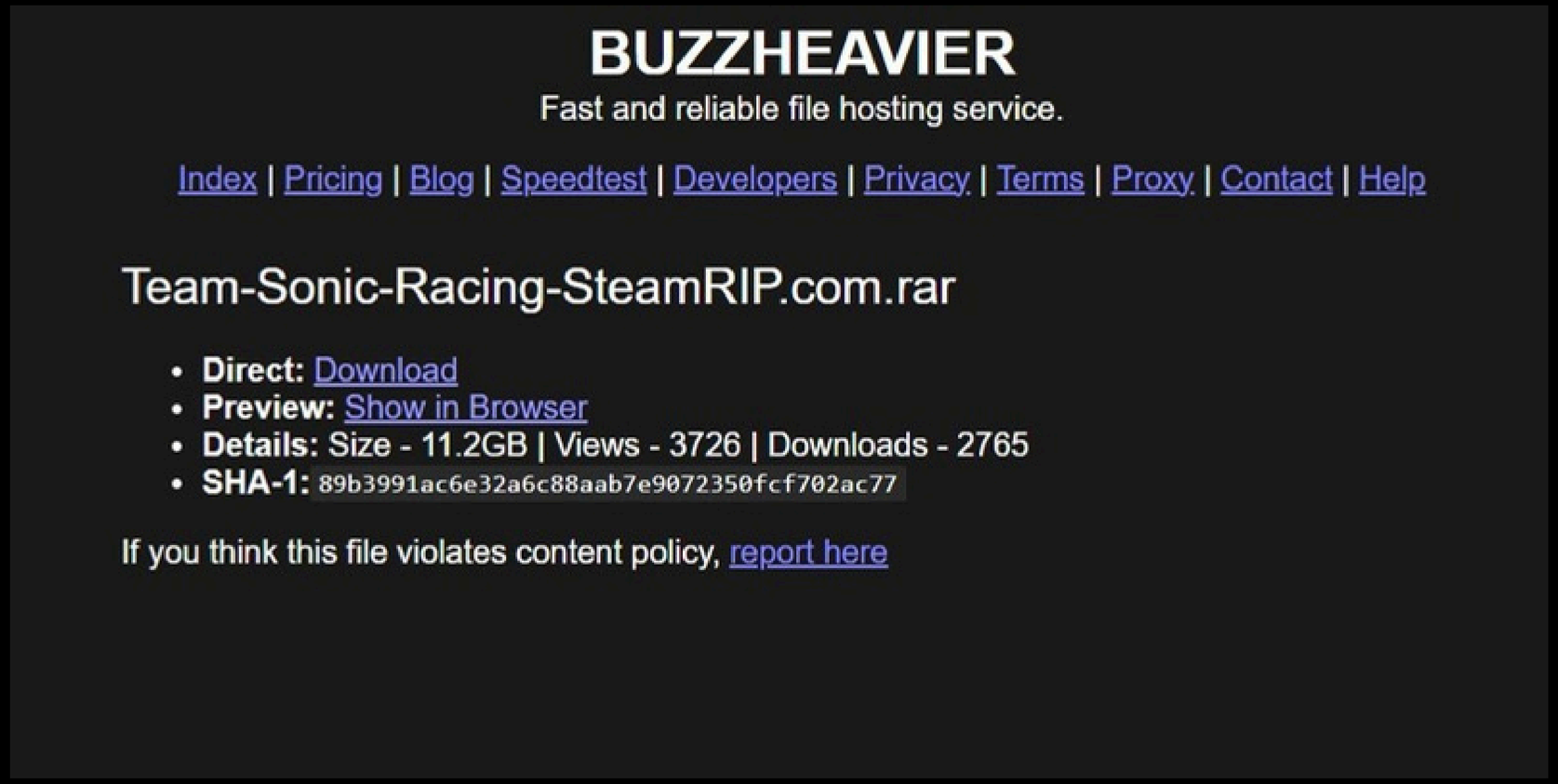
MD5 (Message Digest 5)

- **Output:** 128-bit hash.
- **Speed:** Generally faster, computationally lightweight.

SHA-256 (Secure Hash Algorithm 256-bit)

- **Output:** 256-bit hash.
- **Speed:** Slower than MD5 in older systems, but often faster on modern CPUs with hardware support.

HASH IN ACTION



BUZZHEAVIER

Fast and reliable file hosting service.

Index | Pricing | Blog | Speedtest | Developers | Privacy | Terms | Proxy | Contact | Help

Team-Sonic-Racing-SteamRIP.com.rar

- **Direct:** Download
- **Preview:** Show in Browser
- **Details:** Size - 11.2GB | Views - 3726 | Downloads - 2765
- **SHA-1:** `89b3991ac6e32a6c88aab7e9072350fcf702ac77`

If you think this file violates content policy, report here

HASHCAT

- Hashcat is a high-performance password cracking/recovery tool used in professional security assessments

Advantages

- GPU-accelerated processing
- Supports hundreds of hash algorithms
- Extremely fast compared to CPU-based tools

WLUG
COMMUNITY | KNOWLEDGE | SHARE

MALTEGO

- Maltego is an open-source intelligence (OSINT) and relationship-mapping tool

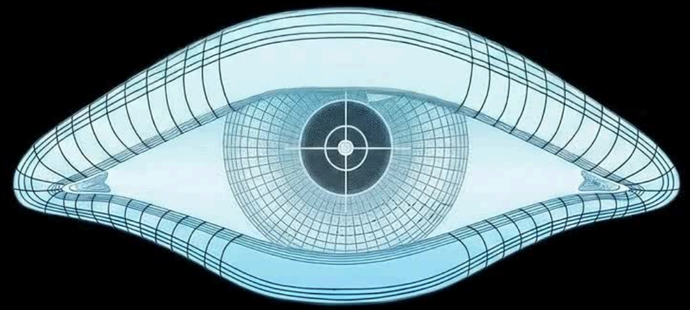
FUNCTIONALITY

- Before hacking begins, attackers gather intelligence
- Maltego shows how much information is already exposed without touching a target system

WLUG
COMMUNITY | KNOWLEDGE | SHARE

NMAP (NETWORK MAPPER)

Nmap is the industry-standard tool for network discovery and security auditing, acting as a 'digital radar' to map out connected devices and identify their open entry points



WLUG
COMMUNITY | KNOWLEDGE | SHARE

CAPABLITIES

- Discover live hosts
- Identify open ports
- Detect running services and versions

IMPORTANCE

- Open ports reveal potential attack paths
- Nmap helps security teams see what an attacker sees

NMAP

```
➜  ~ nmap -T4 scanme.nmap.org
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-01-18 00:18 IST
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.25s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 996 closed tcp ports (conn-refused)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
9929/tcp  open  nping-echo
31337/tcp open  Elite

Nmap done: 1 IP address (1 host up) scanned in 26.74 seconds
```

WLUG
COMMUNITY | KNOWLEDGE | SHARE

WIRESHARK

- Wireshark is a network packet analyzer

WHAT IT DOES ?

- Captures live network traffic
- Decodes protocol behavior
- Identifies insecure communication

WLUG
COMMUNITY | KNOWLEDGE | SHARE

PRECAUTIONS TO BE TAKEN

Cybersecurity is effective only
when good practices are
consistently followed

WLUG
COMMUNITY | KNOWLEDGE | SHARE

PREVENTION

INDIVIDUAL-LEVEL PRECAUTIONS

- Strong, unique passwords
- Multi-factor authentication
- Awareness of phishing tactics

WLUG
COMMUNITY | KNOWLEDGE | SHARE

PREVENTION

ORGANIZATIONAL-LEVEL PRECAUTIONS

- Regular security audits
- Least-privilege access
- Incident response readiness

WLUG
COMMUNITY | KNOWLEDGE | SHARE

CLOSING PERSPECTIVE

CYBERSECURITY IS A CONTINUOUS PROCESS, NOT A ONE-TIME TASK. AS THREATS EVOLVE, SECURITY REMAINS THE FOUNDATION OF A TRUSTWORTHY DIGITAL WORLD—IT IS NO LONGER OPTIONAL, BUT ESSENTIAL.

WLUG
COMMUNITY | KNOWLEDGE | SHARE

PRECAUTIONS & DETECTION

WLUG
COMMUNITY | KNOWLEDGE | SHARE

PRECAUTIONS

- Software updates
- Disable unused services
- Strong password policy

WLUG
COMMUNITY | KNOWLEDGE | SHARE

SOFTWARE UPDATES

WLUG
COMMUNITY | KNOWLEDGE | SHARE

WHICH SEEMS EASY TO BREAK IN?

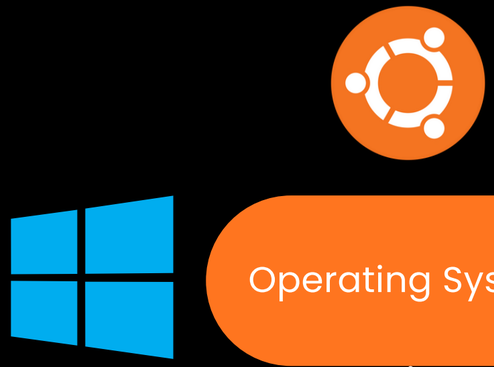
WLUG
COMMUNITY | KNOWLEDGE | SHARE

WHICH SEEMS EASY TO BREAK IN?

v1.8.2

v1.8.6

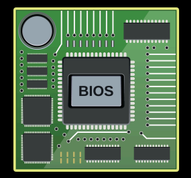
WLUG
COMMUNITY | KNOWLEDGE | SHARE



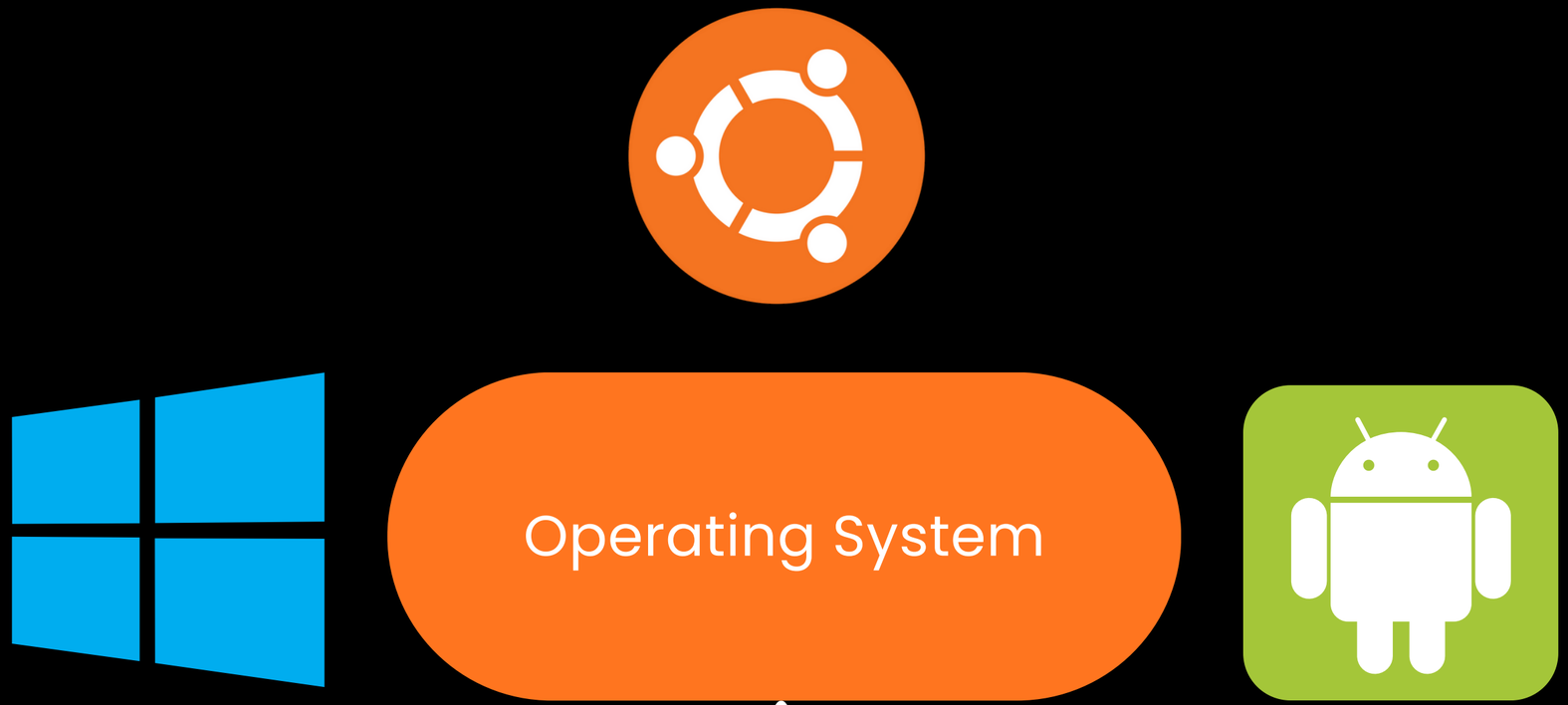
Operating System

Applications

SOFTWARE



Firmware


Operating System

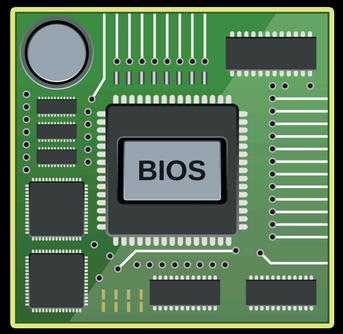
WLUG
COMMUNITY | KNOWLEDGE | SHARE

Applications

SOFT

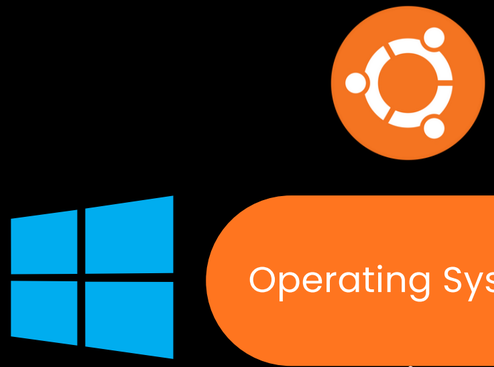
WLUG
COMMUNITY | KNOWLEDGE | SHARE

TWARE



Firmware

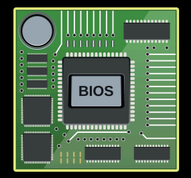
WLUG
COMMUNITY | KNOWLEDGE | SHARE



Operating System

Applications

SOFTWARE



Firmware

WLUG
COMMUNITY | KNOWLEDGE | SHARE

WANNACRY

May-2017

WLUG
COMMUNITY | KNOWLEDGE | SHARE

WANNACRY

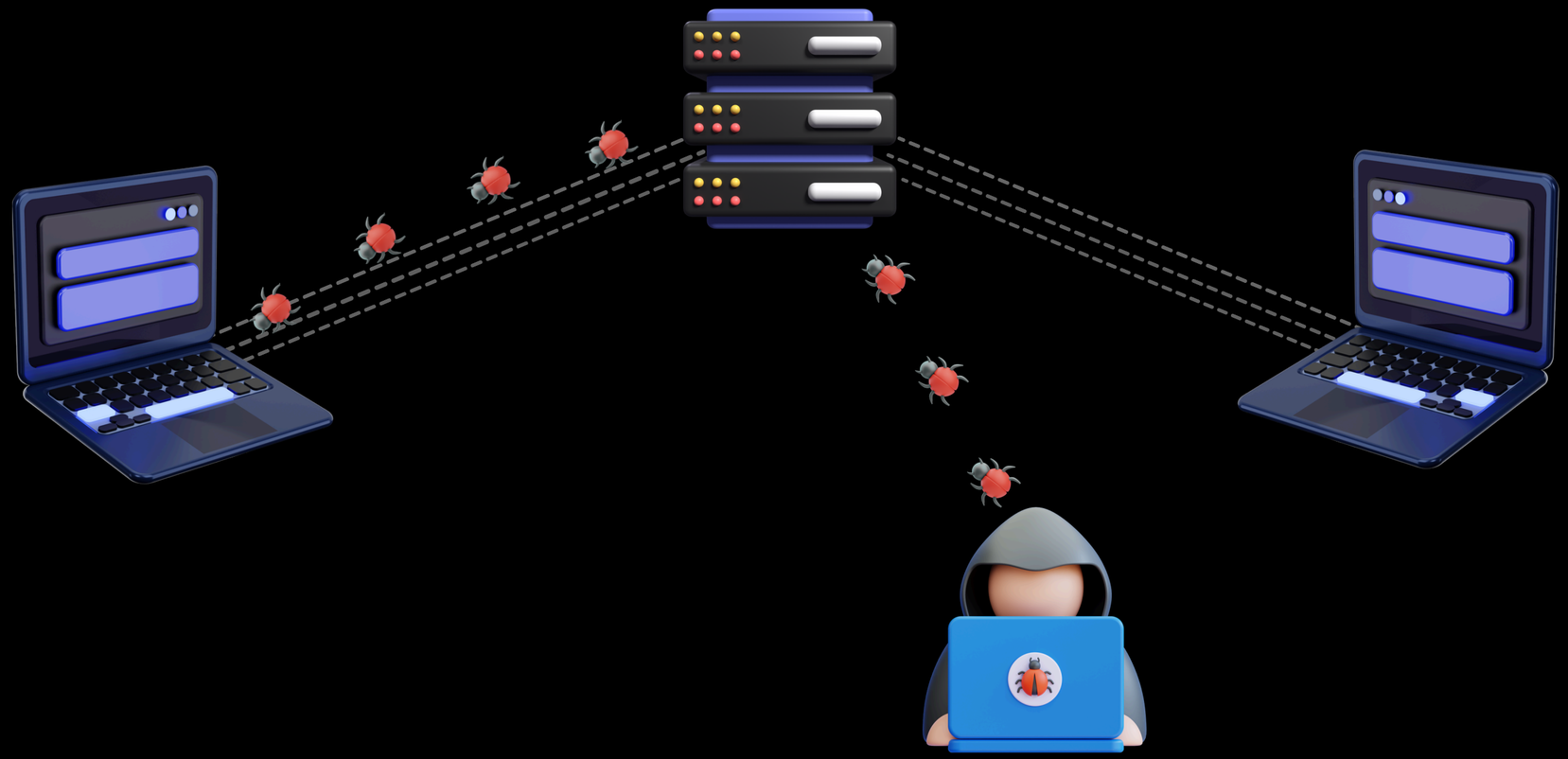
Vulnerability

SMBv1

WLUG
COMMUNITY | KNOWLEDGE | SHARE

SMBv1

Server Message block



WLUG
COMMUNITY | KNOWLEDGE | SHARE

ETERNAL BLUE

WLUG
COMMUNITY | KNOWLEDGE | SHARE

ETERNAL BLUE

WLUG
COMMUNITY | KNOWLEDGE | SHARE

LEAKED

Wana Decrypt0r 2.0
Ooops, your files have been encrypted!
English
What Happened to My Computer?
Your important files are encrypted.
Many of your documents, photos, videos, databases and other files are no longer accessible because they have been encrypted. Maybe you are busy looking for a way to recover your files, but do not waste your time. Nobody can recover your files without our decryption service.
Can I Recover My Files?
Sure. We guarantee that you can recover all your files safely and easily. But you have not so enough time.
You can decrypt some of your files for free. Try now by clicking <Decrypt>.
But if you want to decrypt all your files, you need to pay.
You only have 3 days to submit the payment. After that the price will be doubled.
Also, if you don't pay in 7 days, you won't be able to recover your files forever.
We will have free events for users who are so poor that they couldn't pay in 6 months.
How Do I Pay?
Payment is accepted in Bitcoin only. For more information, click <About bitcoin>.
Please check the current price of Bitcoin and buy some bitcoins. For more information, click <How to buy bitcoins>.
And send the correct amount to the address specified in this window.
After your payment, click <Check Payment>. Best time to check: 9:00am - 11:00am
Payment will be raised on
5/16/2017 00:47:55
Time Left
02:23:57:37
Your files will be lost on
5/20/2017 00:47:55
Time Left
06:23:57:37
About bitcoin
How to buy bitcoins?
Contact Us
bitcoin ACCEPTED HERE
Send $300 worth of bitcoin to this address:
12t9YDPgwueZ9NyMgw519p7AA8isjr6SMw
Copy
Check Payment
Decrypt

WLUG
COMMUNITY | KNOWLEDGE | SHARE

WHY ARE WE DISCUSSING THIS?

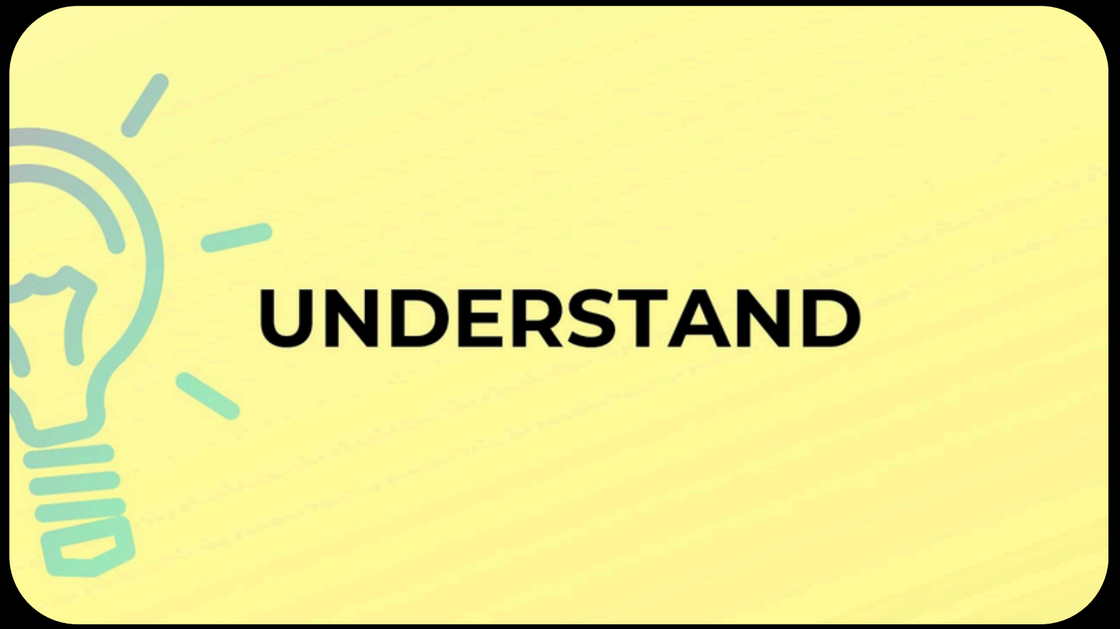
MS17-010: Security update for Windows SMB Server: March 14, 2017

▶ *Applies To*

Summary

This security update resolves vulnerabilities in Microsoft Windows. The most severe of the vulnerabilities could allow remote code execution if an attacker sends specially crafted messages to a Microsoft Server Message Block 1.0 (SMBv1) server.
To learn more about the vulnerability, see Microsoft Security Bulletin MS17-010.

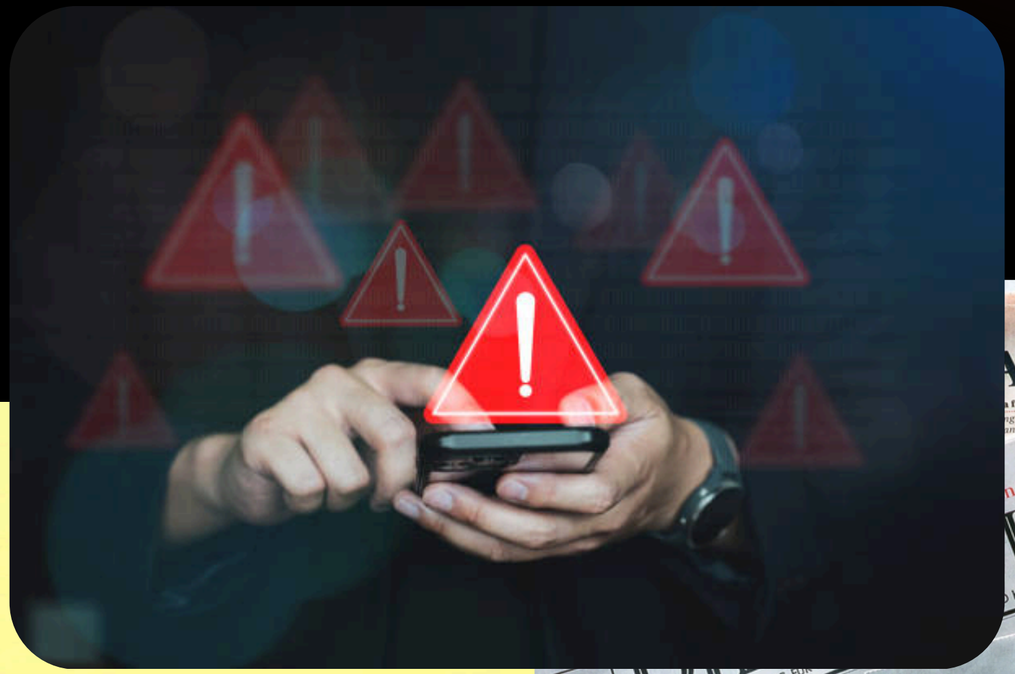
WLUG
COMMUNITY | KNOWLEDGE | SHARE



WLUG
COMMUNITY | KNOWLEDGE | SHARE

UNDE

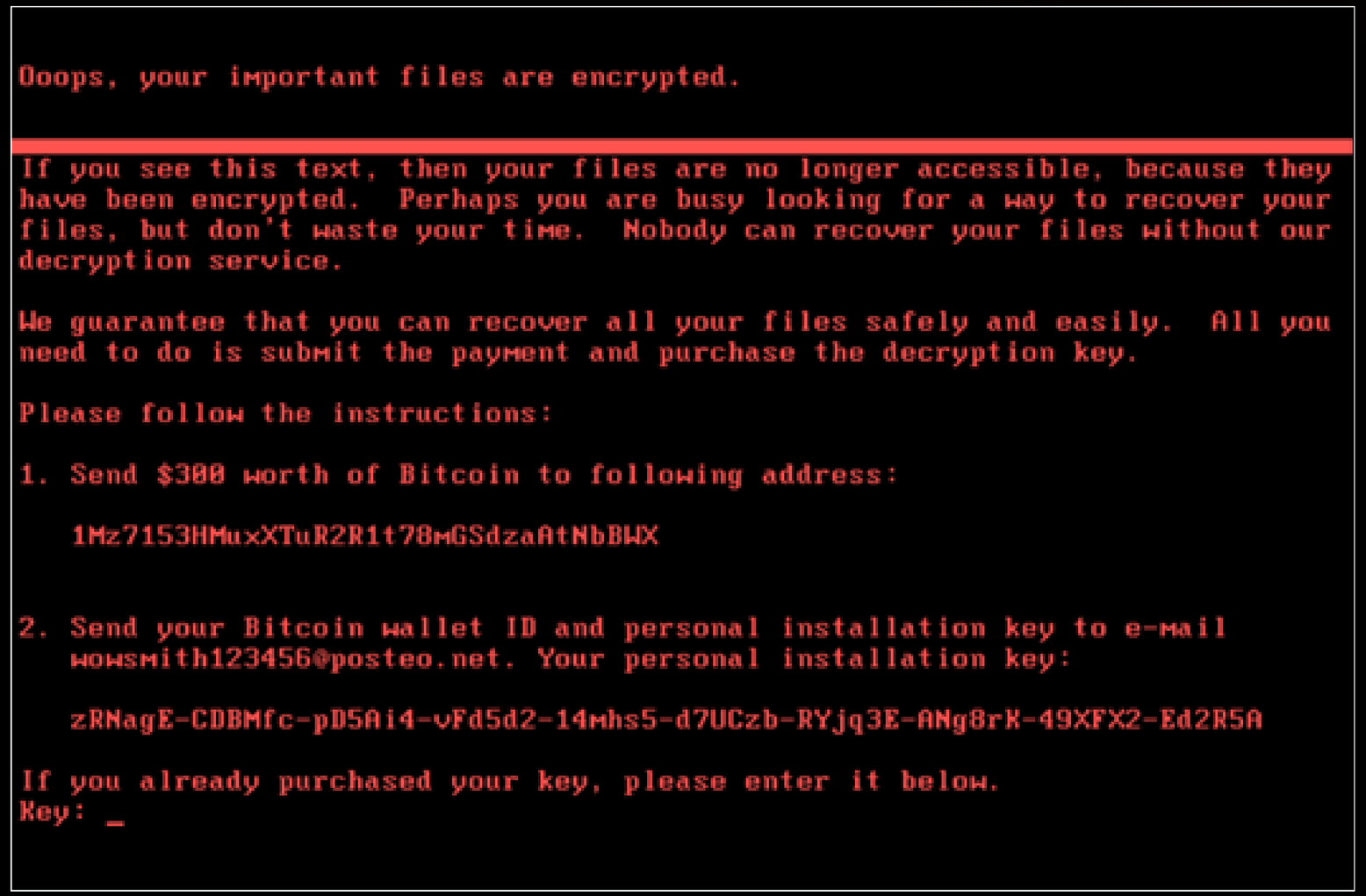
WLUG
COMMUNITY | KNOWLEDGE | SHARE



UNDERSTAND

WLUG
COMMUNITY | KNOWLEDGE | SHARE

June-2017


Ooops, your important files are encrypted.
If you see this text, then your files are no longer accessible, because they have been encrypted. Perhaps you are busy looking for a way to recover your files, but don't waste your time. Nobody can recover your files without our decryption service.
We guarantee that you can recover all your files safely and easily. All you need to do is submit the payment and purchase the decryption key.
Please follow the instructions:
1. Send $300 worth of Bitcoin to following address:
1Mz7153HMuxXTuR2R1t78mGSdzaAtNbBWX
2. Send your Bitcoin wallet ID and personal installation key to e-mail wowsmith123456@posteo.net. Your personal installation key:
zRNagE-CDBMfc-pD5Ai4-vFd5d2-14mhs5-d7UCzb-RYjq3E-ANg8rK-49XFX2-Ed2R5A
If you already purchased your key, please enter it below.
Key: _

WLUG
COMMUNITY | KNOWLEDGE | SHARE

DISABLE UNUSED SERVICES

WLUG
COMMUNITY | KNOWLEDGE | SHARE

WHAT IS A SERVICE?

WLUG
COMMUNITY | KNOWLEDGE | SHARE

WHAT IS A SERVICE?

WLUG
COMMUNITY | KNOWLEDGE | SHARE

WHAT IS A SERVER?

WLUG
COMMUNITY | KNOWLEDGE | SHARE

WHY THIS MATTERS

- SSH service
- Open RDP service
- SMB service
- Database service

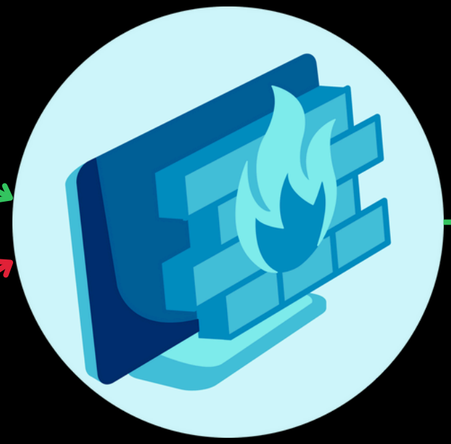
WLUG
COMMUNITY | KNOWLEDGE | SHARE

DISABLING UNNECESSARY SERVICES REDUCES ATTACK SURFACE

WLUG
COMMUNITY | KNOWLEDGE | SHARE

FIREWALL

Network



Your System

Attacker

WLUG
COMMUNITY | KNOWLEDGE | SHARE

STRONG PASSWORD POLICY

WLUG
COMMUNITY | KNOWLEDGE | SHARE

PASSPHRASES OVER PASSWORDS

wlug@osd123

OSD-IS-THE-BEST-EVENT-@180126

WLUG
COMMUNITY | KNOWLEDGE | SHARE

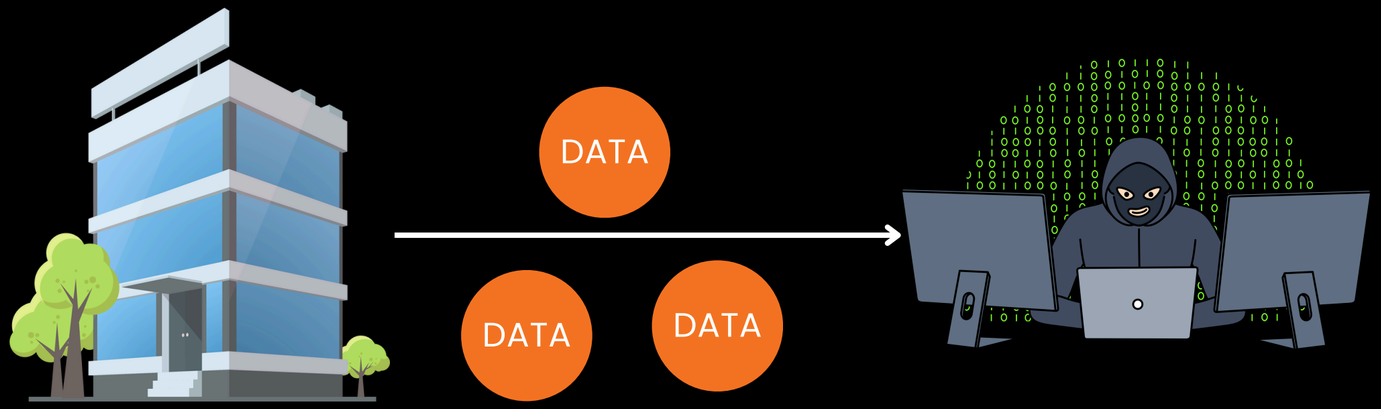
abcdservice.com

Credentials

Username: Neel

Password: neel-says-hi-to-all

abcdservice.com



Credentials

Username: Neel

Password: neel-says-hi-to-all

WLUG
COMMUNITY | KNOWLEDGE | SHARE

abcdservice.com

DATA

DATA

DATA

Social Media

Banking Portal

Email Services

Credentials

Username: Neel

Password: neel-says-hi-to-all

WLUG
COMMUNITY | KNOWLEDGE | SHARE

CREDENTIAL STUFFING

abcdservice.com

DATA

DATA

DATA

Social Media

Banking Portal

Email Services

Credentials

Username: Neel

Password: neel-says-hi-to-all

WLUG
COMMUNITY | KNOWLEDGE | SHARE

GOOD PRACTICES

- Use Multi Factor Authentication
- Passphrases over passwords
- Avoid same credentials

WLUG
COMMUNITY | KNOWLEDGE | SHARE

DETECTION

WLUG
COMMUNITY | KNOWLEDGE | SHARE

DETECTION

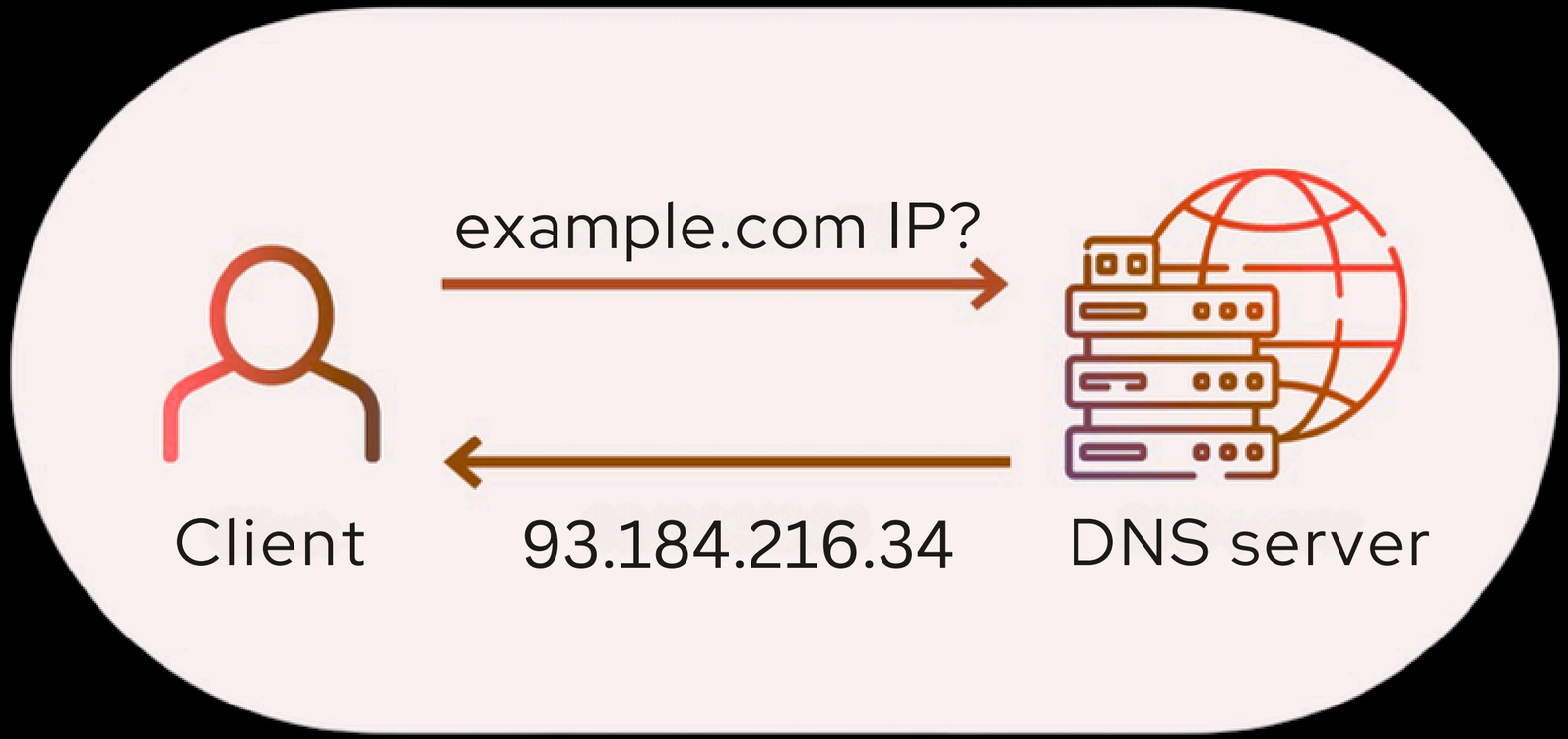
- Unusal DNS queries
- Large outbound traffic
- Repeated failed auth attempts
- Log analysis

WLUG
COMMUNITY | KNOWLEDGE | SHARE

UNUSUAL DNS QUERIES

WLUG
COMMUNITY | KNOWLEDGE | SHARE

DOMAIN NAME SYSTEM



WLUG
COMMUNITY | KNOWLEDGE | SHARE

DOMAIN NAME SYSTEM

Usual DNS queries

DNS: youtube.com

- Domains are well-known and readable
- Happens occasionally, only when needed

WLUG
COMMUNITY | KNOWLEDGE | SHARE

DOMAIN NAME SYSTEM

Unusual DNS queries

DNS: x9a82kd92kdk.attacker-domain.com

- Random-looking domain names
- DNS tunneling

WLUG
COMMUNITY | KNOWLEDGE | SHARE

SOLUTION

- DNS Filtering & Secure DNS Resolvers
  - Cloudflare (1.1.1.1)
  - Google (8.8.8.8)

- Block Newly Registered & Rare Domains
  - Malware frequently uses fresh domains
  - Block domains registered in last 7-10 days

WLUG
COMMUNITY | KNOWLEDGE | SHARE

LARGE OUTBOUND TRAFFIC

WLUG
COMMUNITY | KNOWLEDGE | SHARE

NORMAL BEHAVIOR:

- Predictable patterns (daytime peaks)
- Higher usage during office / active hours

ABNORMAL BEHAVIOR:

- No users logged in
- Huge data upload at midnight / early morning
- Possible data exfiltration

DATA EXFILTRATION

- Source code
- Credentials
- Customer data

WHY AT ODD HOURS?

- Security teams less active
- Less monitoring

3:17 AM

UPLOAD 5GB

Destination IP

WLUG
COMMUNITY | KNOWLEDGE | SHARE

SOLUTION

Immediate Actions

- Isolate the affected system
- Capture traffic for analysis
- Block destination IP/domain

Long-term Controls

- Egress firewall filtering
- Data Loss Prevention (DLP)
- Alerting on bandwidth spikes

WLUG
COMMUNITY | KNOWLEDGE | SHARE

REPEATED FAILED AUTH ATTEMPTS

WLUG
COMMUNITY | KNOWLEDGE | SHARE

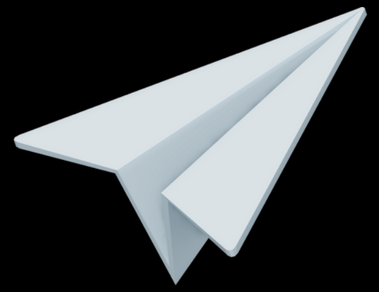
BRUTE FORCE ATTACK

- Attacker tries thousands of password combinations
- Common on:
  - SSH
  - RDP
  - Web login pages

WLUG
COMMUNITY | KNOWLEDGE | SHARE

SOLUTION

- Limit how many login attempts are allowed
- Automatically block IPs after failures
- Tools
  - Fail2Ban (Linux)
  - Web server rate limits (Nginx, Apache)



WLUG
COMMUNITY | KNOWLEDGE | SHARE

NEW LISTENING PORT

WLUG
COMMUNITY | KNOWLEDGE | SHARE

NEW LISTENING PORT:

- A port unexpectedly appears open
- No admin intentionally enabled it
- Maybe used as a Backdoor

WLUG
COMMUNITY | KNOWLEDGE | SHARE

XZ UTILS BACKDOOR

WLUG
COMMUNITY | KNOWLEDGE | SHARE

COMPRESSION

10 MB

1 MB

WLUG
COMMUNITY | KNOWLEDGE | SHARE

Lasse Collin
(Maintainer)

XZ UTILS 5.6.0

WLUG
COMMUNITY | KNOWLEDGE | SHARE

Lasse Collin
(Maintainer)

JIA TAN
(Contributor)

XZ UTILS 5.6.0

WLUG
COMMUNITY | KNOWLEDGE | SHARE

Lasse Collin
(Maintainer)

JIA TAN
(Contributor)

XZ UTILS 5.6.0

WLUG
COMMUNITY | KNOWLEDGE | SHARE

Lasse Collin
(Maintainer)

Andres Freund

JIA TAN
(Contributor)

XZ UTILS 5.6.0

WLUG
COMMUNITY | KNOWLEDGE | SHARE

SOLUTION

Immediate Actions

- Identify process owning the port
- Block port via firewall
- Kill the process

Restore System Integrity

- Restore from clean backup
- Reinstall OS (recommended for serious compromise)
- Remove malware manually

WLUG
COMMUNITY | KNOWLEDGE | SHARE

LOG ANALYSIS

WLUG
COMMUNITY | KNOWLEDGE | SHARE

LOGS

- Logs are automatic records created by systems when something happens
- What it contains
  - Login attempts
  - Service start / stop
  - Network connections
  - User creation

WLUG
COMMUNITY | KNOWLEDGE | SHARE

RED FLAGS

- Repeated access to sensitive files
- New admin account creation
- Many failed logins from one IP

WLUG
COMMUNITY | KNOWLEDGE | SHARE

THANK YOU!